GraspGrade: Scoring First-Person Video Clips for Robot Hand Imitation, Checked Against Touch Gloves and Motion Capture

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Abstract

Robot hand teams want first-person human video because it is cheap and plentiful, but most of it cannot teach a robot hand. Every team writes its own filter, and few of those filters are checked against a sensor. GraspGrade cuts first-person video into clips at the moment of contact and gives each clip a score for how likely it is that a robot hand can copy the grasp. Video is the only input. Two sensor sources are used only to score the work: a touch-sensing glove for contact timing, and motion capture of hands and objects for robot copyability. On 22 held-out glove episodes, a vision language model asked directly about touch, plus one timing shift learned from the glove on tuning episodes, gives the best contact timing of the arms we compare (boundary F1 0.195 at $\pm$5 frames, 0.376 at $\pm$15 frames, contact IoU 0.850). On 1,426 held-out motion capture clips, the same model asked the plain question "can a robot hand copy this?" is at chance (AUC 0.501), while a fixed checklist of object and hand questions reaches 0.624. The gain comes from the checklist (+0.117), not from learning weights on labels (+0.005). The name of the object alone reaches 0.771 when the object was labeled before, and our score keeps 0.571 on objects it has never seen, where the object name has nothing to say. A step meant to hold soft objects back misses its pre-registered rule. Both ground truths are released as label files with standalone scorers, so other filters can be scored against the same sensors. Every comparison, threshold and pass rule was written down before the held-out data was scored.

1 Introduction

A robot hand that learns from human video needs clips where the hand is visible, the touch is clear, and the grasp is one the robot hand can actually reproduce. Most internet video fails this. In a recent study only 83 of 2,000 clips, about 4 percent, passed a quality filter [1]. Teams usually describe the reasons they drop a clip, but they rarely publish a score that another team can reuse, and almost never one that is checked against a sensor rather than against human opinion.

Two questions have to be answered before any clip is useful.

1. **When does a touch start and end?** The clip has to be cut at the right frames. From video alone this is hard, because pressure is not visible.
2. **Can a robot hand copy this grasp?** A human hand reaches places a four-finger robot hand cannot, and a robot finger is thicker than a human finger.

The common option today is to show a vision language model a clip and ask it whether a robot hand can copy the grasp. On our tuning data this is close to chance.

GraspGrade answers both questions from video alone and checks each answer against a sensor, not against our own judgement.

Contributions.

- A contact cutting comparison on a touch-sensing glove dataset, with three arms built from Meta SAM 3.1 and Muse Spark, and a timing shift learned from the glove. We report the full table, including the rows our own pipeline loses (Section 5).
- A copyable label for grasp clips, built from motion capture hands and objects and a position retargeting of the human hand onto an Allegro robot hand, with mesh collision and contact reach conditions. The label passes an acceptance rule fixed before it was computed, and we report it at three collision depth limits (Section 6).
- A video-only score from twelve graded checks, and six comparisons that separate what the checklist adds from what learning adds, and what the object name alone already tells. The score is tested on objects it has never seen (Section 7).
- A negative result, reported as it came out: a soft object step that fails its pre-registered pass rule (Section 8).
- A benchmark release of both ground truths as label files with standalone scorers that reproduce our numbers exactly (Section 9).

Everything was run under a pre-registration discipline. Every threshold, prompt and weight was chosen on a tuning split, and every reported number comes from a held-out split that was scored once.

2 Related work

Recent work on dexterous manipulation from human video filters clips before retargeting them. "Do as I Do" reports that 83 of 2,000 everyday human video clips survive its quality check [1]. SPIDER [2] and ManipTrans [3] retarget human hand motion to robot hands with physics in the loop, which is accurate and expensive per clip. GraspGrade runs before that step. It is a cheap geometric filter that decides which clips are worth the expensive treatment, and it is scored against sensors.

For the models and data we use Meta SAM 3.1 for hand and object tracking, Meta Muse Spark as the vision language model, HOT3D [4] for hand and object motion capture, OpenGraph Ego-Tactile [5] for tactile gloves, and dex-retargeting [6] for the human to robot hand mapping.

3 Data and splits

OpenGraph Ego-Tactile. 72 episodes of first-person video at 30 fps, 138,629 frames, 1.28 hours. Both hands wear a glove with 80 taxels. The camera is an iPhone ultra-wide, about 107 degrees horizontal field of view. Splits: 50 tuning episodes, 22 held-out episodes.

HOT3D (Project Aria). 136 Aria sequences from 9 persons have hand and object pose ground truth, 480,375 RGB frames at 30 fps, fisheye, 1408 by 1408. There are 33 objects with 3D models. Splits by person: 6 persons and 85 sequences for tuning, 3 persons and 51 sequences held out.

Both splits were drawn once with a fixed seed and never re-rolled. Neither dataset is redistributed here.

Clips. A grasp clip is cut from motion capture: two fingertips within 15 mm of an object, one second added before and after, the object moved at least 3 cm, and the clip passes the dataset quality check. One clip belongs to one hand. This gives 2,385 tuning clips and 1,426 held-out clips.

Ground truth for contact timing. A hand is in contact when the sum of its 80 taxels is above 800. Pressure dropouts of 0.1 s or less are joined and pressure blips shorter than 0.1 s are dropped, because that flicker is threshold noise no video method can see. The raw glove and the dataset contact labels are reported next to it as checks. The three agree on about 97 to 99 percent of frames.

4 Definitions and thresholds

Every rule below was fixed on the tuning split before the held-out split was scored. The experiment id after each value says where it was fixed.

Task 1 ground truth, the cleaned glove.

$$c_t = \mathbf{1}\Big[\textstyle\sum_{i=1}^{80} p_{i,t} > 800\Big] \tag{1}$$

where $p_{i,t}$ is the pressure of taxel i of one hand at frame t, at 30 fps. Contacts broken for 3 frames (0.1 s) or less are joined, and pieces shorter than 0.1 s are dropped (EXP-04, EXP-14).

Task 1, arm B, the SAM mask rule at 10 fps.

$$near_t = \mathbf{1}[\,d_t \le \tau_d\,] \tag{2}$$

$$m_t = \max\big(\|\Delta o_{t-3\to t}\|,\ \|\Delta o_{t\to t+3}\|\big)/\sqrt{A_o} \tag{3}$$

$$\kappa_t = \frac{\|\Delta o - \Delta h\|}{\|\Delta o\| + \|\Delta h\| + 5} \tag{4}$$

$$move_t = near_t \wedge hand_t \wedge (m_t \ge \tau_m) \wedge (\kappa_t \le \tau_\kappa) \tag{5}$$

Here d_t is the pixel distance between the SAM hand mask and the nearest object mask, Δo and Δh are the motions of the object and hand mask centres with camera motion removed, camera motion is the median motion of the other objects in the same window, and A_o is the median object mask area. So m_t is object motion in units of object size, and κ_t is near zero when hand and object move together. A contact segment is a run of $near_t$ that holds at least one $move_t$ frame. Gaps of f frames or less are joined and segments shorter than 0.2 s are dropped. A segment is also cut when the nearest object changes and stays changed for q frames (EXP-22, EXP-24).

The thresholds come from a grid search on the tuning episodes, $\arg\max[\mathrm{F1}_{\pm 15} + \mathrm{IoU}]$ over $\tau_d \in \{2,4,8,12\}$ px, $\tau_m \in \{0.05, 0.1, 0.2\}$, $\tau_\kappa \in \{0.5, 0.7\}$, $f \in \{0,1,2\}$ and $q \in \{1,3,5\}$. The 50 tuning episodes chose $\tau_d = 2$ px, $\tau_m = 0.05$, $\tau_\kappa = 0.7$, $f = 0$, $q = 3$, with shifts $\delta_s = +4$ and $\delta_e = -1$ frames (EXP-32).

Task 1 timing shift, the same form for every arm.

$$\delta_s = -\,\mathrm{med}_j(\hat{s}_j - s_j), \qquad \delta_e = -\,\mathrm{med}_j(\hat{e}_j - e_j) \tag{6}$$

$\hat{s}$ and $\hat{e}$ are the predicted start and end, s and e the glove start and end, matched one to one within 15 frames on tuning episodes only. Arm A gets $\delta_s = +7$ and $\delta_e = -7$ frames (EXP-29). Arm C learns only from the tuning boundaries that Muse Spark moved, and gets $+6$ and -4 (EXP-36).

Task 1 metrics. Predicted and glove boundaries are matched one to one within $|\hat{s} - s| \le k$, each only to its own kind, with $k = 5$ and $k = 15$ frames (0.167 s and 0.5 s). Then P is matches over predicted boundaries, R is matches over

glove boundaries, $\mathrm{F1}_{\pm k} = 2PR/(P+R)$, and the contact IoU is the overlap of predicted and glove contact frames over their union.

Task 2 ground truth, the copyable version 2 label.

$$y = \mathbf{1}\left[\exists u \in U:\ \mathrm{grip}(u) \wedge \mathrm{reach}(u) \wedge \neg\,\mathrm{collide}(u)\right] \quad (7)$$

U is a shift of the whole robot hand on a 1 cm grid up to 2 cm. The three conditions are given in Section 6, and all thresholds were fixed before computing (EXP-16).

Task 2 grades.

$$g(x) = \begin{cases} 1 & x \geq q_{2/3} \\ 0.5 & q_{1/3} \leq x < q_{2/3} \\ 0 & x < q_{1/3} \end{cases} \quad (8)$$

with q the terciles of the tuning clips. The raw values x are listed in Section 7.

Task 2 score and the rows it is compared with.

$$S = 100\,\sigma\left(w_0 + \textstyle\sum_{i=1}^{12} w_i g_i\right) \quad (9)$$

$$R_2 = \tfrac{1}{7}\textstyle\sum_{i \in \mathrm{Muse}} g_i \quad (10)$$

$$R_3 = \sigma\left(w_0 + \textstyle\sum_{i \in \mathrm{Muse}} w_i g_i\right) \quad (11)$$

with $\sigma(z) = 1/(1+e^{-z})$. R_1 is the copyable probability that Muse Spark gives when it is asked the question directly, and $r(o)$ is the copyable share of object o on the tuning clips. The weights w come from an L2 logistic regression ($C = 0.5$) on the 400 tuning clips. AUC is $P(S^+ > S^-) + \frac{1}{2}P(S^+ = S^-)$, with a 95 percent interval from a bootstrap over held-out sequences. For the unseen object rows, the hand shape model, the terciles and the weights are all refit without the object being scored, so $r(o)$ gives every clip the same value and sits at AUC 0.5.

5 Task 1: when does a touch start and end?

5.1 Three arms

All three arms take video only and output contact segments per hand at 30 fps.

A. Muse Spark alone. The model sees frames at 5 fps, several frames per call with their times, and says per frame and per hand whether the hand touches an object. No SAM.

B. SAM rule. SAM 3.1 tracks masks for the prompts `hand` and `small object` at 10 fps. Contact comes from mask distance plus co-movement, that is, the object moving with the hand after camera motion is removed (Section 4). The thresholds were chosen on the 50 tuning episodes by leaving one episode out, and 46 of 50 folds picked the same rule.

C. SAM rule plus a Muse Spark boundary check. For every start and end that arm B produces, Muse Spark sees about 7 frames around it at 30 fps and picks the first or last touch frame, with a reason and a confidence.

Two reference rows make the numbers readable: "always in contact" and "never in contact".

Table 1: Task 1 on the 22 held-out episodes, 44 episode-hands, 574 cleaned glove boundaries (EXP-37). The best arm is in bold. The last column is the number of boundaries the arm produces.

Row	F1 ±5	F1 ±15	IoU	Bounds
A. Muse Spark alone	0.074	0.273	0.821	400
A + shift	**0.195**	**0.376**	**0.850**	358
B. SAM rule	0.090	0.214	0.667	1,790
C. B + Muse Spark check	0.112	0.227	0.679	1,646
C + shift	0.137	0.266	0.690	1,546
Always in contact	0.042	0.042	0.600	88
Never in contact	0	0	0	0
Glove (ground truth)				574

5.2 Timing shift from the glove

Video shows a touch before the glove feels pressure, by about 0.2 s. Judged against the glove with no correction, every video arm looks late at the start and early at the end. Each Muse Spark arm therefore gets one constant shift per boundary kind, learned on tuning episodes only (Section 4). Arm A gets $+7$ frames at the start and -7 at the end. Arm C gets $+6$ and -4, measured only on the boundaries that Muse Spark moved. The shift is two numbers per arm. It is a calibration from sensor truth and not a new model, and the unshifted row is reported next to it.

5.3 Metrics

Boundary F1 at ±5 frames and at ±15 frames, and contact IoU over all frames (Section 4), reported pooled and per hand. The test for the shift is a paired bootstrap over the 22 held-out episodes, 1,000 draws, F1 at ±5 frames. The shift counts as useful only if the 95 percent interval of the difference lies above zero.

5.4 Results

Table 1 and Figure 1 give the result. Muse Spark alone with the glove shift is the best row on all three metrics. The shift is useful for both Muse Spark arms by the pre-registered test (Table 2).

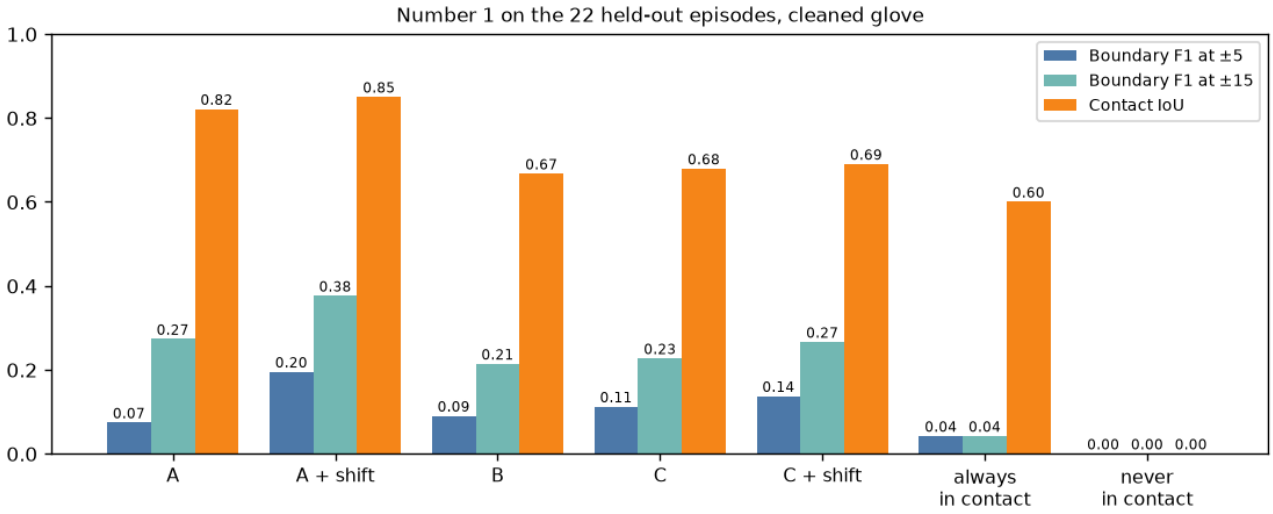


Figure 1: Task 1 on the 22 held-out episodes, cleaned glove ground truth (EXP-37).

Table 2: Paired bootstrap over the 22 held-out episodes, F1 at ±5 frames. The first two rows were written in the plan before the run. The rest are descriptive, added after it (EXP-37).

Difference	Value	95% interval
(A + shift) − A	+0.121	0.062 to 0.172
(C + shift) − C	+0.025	0.019 to 0.032
C − B, ±5	+0.022	0.004 to 0.037
C − B, ±15	+0.013	0.002 to 0.023
(A + shift) − (C + shift), ±5	+0.058	−0.017 to 0.119
(A + shift) − (C + shift), ±15	+0.109	0.020 to 0.221

The Muse Spark boundary check does improve the SAM rule, by +0.022 at ±5 frames, but arm C stays below arm A with the shift. The reason is visible in the boundary counts: the SAM rule produces 1,790 boundaries where the glove has 574, about three times too many, and arm C can only move boundaries, never remove them. Arm B also drops from tuning to held-out, from F1 0.153 to 0.090 at ±5 frames.

All rows are low at ±5 frames, which the plan expected. The glove measures pressure, and pressure is not visible. At ±15 frames, half a second, arm A with the shift finds 0.376 of the boundaries, and its contact IoU of 0.850 is well above the 0.600 of the always in contact row.

Per hand, arm A with the shift reaches F1 0.246 at ±5 frames on the left hand and 0.159 on the right. In 9 short episodes where arms A and B disagreed strongly, the glove puts the left hand in contact in only 9.4 percent of frames, and Muse Spark was right: arm A with the shift reaches IoU 0.835 there, while arm B makes 402 boundaries for 20 real ones and reaches IoU 0.104.

6 A copyable label from motion capture

Task 2 needs a ground truth for "a robot hand can copy this grasp". We build it from HOT3D motion capture, with no video involved.

6.1 First attempt and why it failed

Our first label scaled the wrist to fingertip vectors by the ratio of robot to human hand size, and checked collisions at link origin points only. It gave 54 percent copyable, but a check written before computing showed it mostly measured a retargeting side effect. Robot fingertips floated about 25 mm off cans that the person held firmly, and the copyable share per person tracked hand length, from 29 to 74 percent. Faithful retargeting instead called 94 to 96 percent of grasps copyable, because origin-point collisions miss the thick robot fingers. No method passed the acceptance rule, so the label was put on hold and rebuilt.

6.2 Copyable, version 2

The thresholds and the acceptance rule were written down before computing.

For each clip we take the contact frame and the clip's hand. The human hand is mapped onto an Allegro hand by position retargeting, which fits Allegro fingertips and finger bases to the human ones with a free hand base. The Allegro surface is about 300 sampled points on the visual mesh of each of its 21 links, placed by forward kinematics. The hand may shift up to 2 cm on a 1 cm grid. A clip is copyable if one shift passes all three conditions:

1. **Grip.** The Allegro thumb tip and one more Allegro fingertip are within 1.5 cm of the object surface.
2. **Reach.** Every human finger that touches the object, that is a tip within 15 mm, has its Allegro fingertip within 2.0 cm of that human fingertip.
3. **No collision.** No Allegro mesh point is more than 5 mm inside the object.

The acceptance rule, fixed in advance, asked for a copyable share between 20 and 80 percent and a rank correlation of hand length with the label inside objects of at most 0.1. The label passes both: 47.0 percent copyable on tuning clips, and a hand size correlation of $+0.04$, with 44 to 54 percent by person. On held-out clips the same fixed rule gives 46.4 percent, and the per-object shares agree between the splits with a rank correlation of 0.85.

Failures make physical sense. Collision is the binding condition for 25.5 percent of clips, reach for 10.9 percent and grip for 3.9 percent. The thicker Allegro fingers go into flat or crowded objects such as a keyboard or a spatula, and the hand cannot span a wide bowl or plate the way the human did. By object, the share runs from 1 percent for a keyboard to 96 percent for a small holder.

The main weakness of the label. The share depends strongly on the collision depth limit: 18 percent at 2 mm, 47 percent at 5 mm and 66 percent at 10 mm (Figure 2). The 5 mm limit is about the size of the motion capture and model error, so the label is partly a statement about how much the robot hand overlaps the object within measurement error. The limit was fixed before computing, we report the whole curve, and Section 7.4 scores the held-out result at 2 mm and 10 mm as well.

7 Task 2: does the score pick clips a robot hand can copy?

7.1 The score

Each clip gets twelve graded checks. Each grade is high, mid or low, coded 1, 0.5 and 0. A logistic regression on 400 tuning clips turns them into a score from 0 to 100. We use a logistic regression and not a decision tree, so that every weight is readable and every check contributes in a fixed direction.

Three local checks, no API call. Camera stability: minus the 90th percentile of camera motion over the clip, from corner tracking and a RANSAC similarity fit between frames. Hand shape fit: the probability from a logistic regression on ten MediaPipe hand features during the contact, trained on tuning clips. Action completeness, a fixed rule and not a tercile: half the sum of two indicators, one for the second before contact and one for the second after it, each true when the hand is seen in at least 50 percent of frames and the wrist moves at least 3 percent of the frame width. Checks of the checks: camera motion against headset rotation from motion capture correlates at 0.93 (Spearman), and the MediaPipe hand label matches the motion capture hand in 96.5 percent of frames (EXP-12).

Two SAM 3.1 checks. Visibility: the share of frames where SAM found both the hand and the object. Contact clarity: the share of contact frames with $d_t \leq 3$ px, minus the same share over the reach and release frames. Both use the masks of the prompts `hand` and `small object` at 10 fps, and both are graded by the tuning terciles. Visibility turned out to have only two effective grades, because about 70 percent of clips have full visibility (EXP-38).

Seven Muse Spark grades. Six object items plus finger occlusion. The object items are size next to the hand, shape, thickness, width the hand must wrap, handle and free space around the object. High always means easier for a large four-finger robot hand, and the anchors use the visible human hand as the ruler, about 19 cm from wrist to middle fingertip.

The object items were settled by two free checks before any paid call. First, whether object geometry can carry the signal that the object name carries. Seven size and shape numbers taken from the 3D models reach AUC 0.695 when a person is left out and 0.673 when the object itself is left out, against 0.742 and 0.500 for the object name (EXP-18, Figure 4). Second, whether coarse three-level stand-in grades keep that signal: they reach 0.683 and 0.666 (EXP-19). Both checks had to clear a 0.65 bar set in advance, and both did.

7.2 What is compared

Six comparisons, all written down before the held-out run.

1. Our score against Muse Spark asked the same question with no checklist.
2. An object name reference: the copyable share of each object on the tuning clips, handed to each held-out clip. It needs a label for every object, so it is a reference and not a video method.
3. Muse Spark alone against Muse Spark plus our checks, that is GraspGrade as a layer on an existing filter.
4. Comparisons 1 and 2 again on objects the score has never seen, with every learned part refit without the object being scored.
5. Three rows on the same clips: Muse Spark alone, the checklist with equal weights, and the checklist with learned weights. This separates what the checklist adds from what learning adds.
6. Our score with the object rate added when 0, 5, 20 or all labeled tuning clips of the object exist.

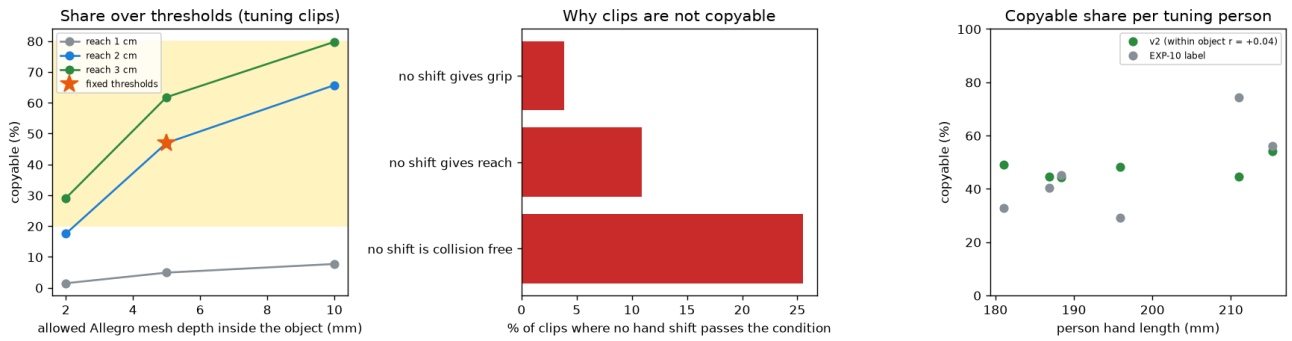


Figure 2: The copyable version 2 label. Left: the share for other reach and depth limits, with the fixed choice starred. Middle: which condition no hand shift can meet. Right: the share per tuning person for version 2 (green) and for the first label (grey). EXP-16.

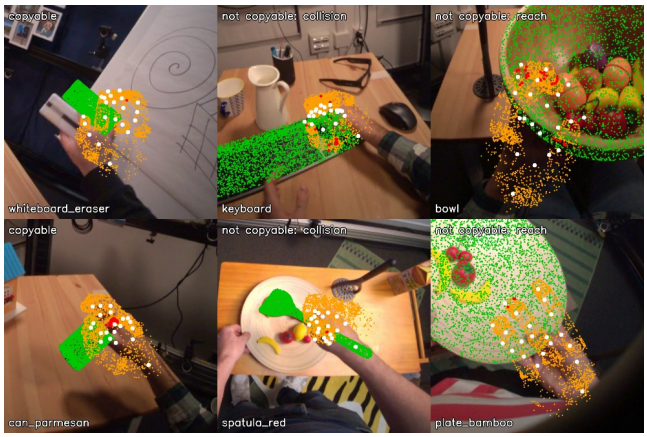


Figure 3: Two tuning clips per outcome. Green is the object model, white the human hand, orange the Allegro mesh points and red the points more than 5 mm inside the object. Left: copyable. Middle: collision. Right: reach (EXP-16).

AUC is reported with a 95 percent bootstrap interval that resamples the 51 held-out sequences, not single clips, because clips from one sequence are related. Paired differences use the same sequences for both rows.

7.3 Results

The checklist is what works, not learning. Muse Spark asked the plain question cannot tell copyable from non-copyable clips at all (AUC 0.501, Table 3). The same grades averaged with no learning at all reach 0.617. Learning the weights on 400 labeled clips adds +0.005, and the interval does not separate that from zero. The local and SAM checks add +0.001 on top (Table 4). So the useful part is the question set, not the labeled training data, and the method can be used without labels.

The score is stable across the three held-out persons, at 0.603 to 0.638, and 59.4 percent of the clips in its top fifth are copyable against a base rate of 46.4 percent. Used as a layer on top of Muse Spark alone it gives the same 0.624.

Table 3: Task 2 on the 1,426 held-out clips, 3 persons, 46.4 percent copyable. "Top 20%" is the share of copyable clips in the top fifth of the ranking (EXP-40).

Row	AUC	95% interval	Top 20%
Our score	**0.624**	0.589 to 0.661	59.4%
1. Muse Spark alone	0.501	0.469 to 0.529	46.9%
2. Object name reference	0.771	0.744 to 0.792	79.4%
3. Muse alone + our checks	0.624	0.590 to 0.661	62.6%
Comparison 4, objects never seen in training			
Our score	0.571	0.534 to 0.609	
Object name reference	0.500	by construction	
Muse Spark alone	0.501	0.469 to 0.529	

The object name beats our score. The copyable share of the object, taken from tuning labels, reaches 0.771. That is the strongest row in the table, and our score loses to it. It is a reference and not a video method, because the held-out persons grasp the same 33 objects as the tuning persons, and because a new object has no share. When every learned part is refit without the object being scored, our score keeps 0.571 and the object reference drops to 0.500 by construction. Video of objects nobody has measured is the case the score is built for.

The score ranks objects, not grasps. Inside a single object, over the 13,524 clip pairs that share an object, our score is at chance (0.506, interval 0.461 to 0.544). Comparison 6 says the same from the other side: once 5 or more labeled clips of an object exist, the object rate alone is as good as the combination, and adding our score to it gains nothing (+0.004 at 5 clips, −0.003 with all clips). The object rate added to our score gains +0.099 to +0.147 in the other direction. The object rate is the stronger of the two wherever it exists.

Which failures it sees. Against clips that fail only on grip, the score reaches 0.682. Against reach failures, 0.636.

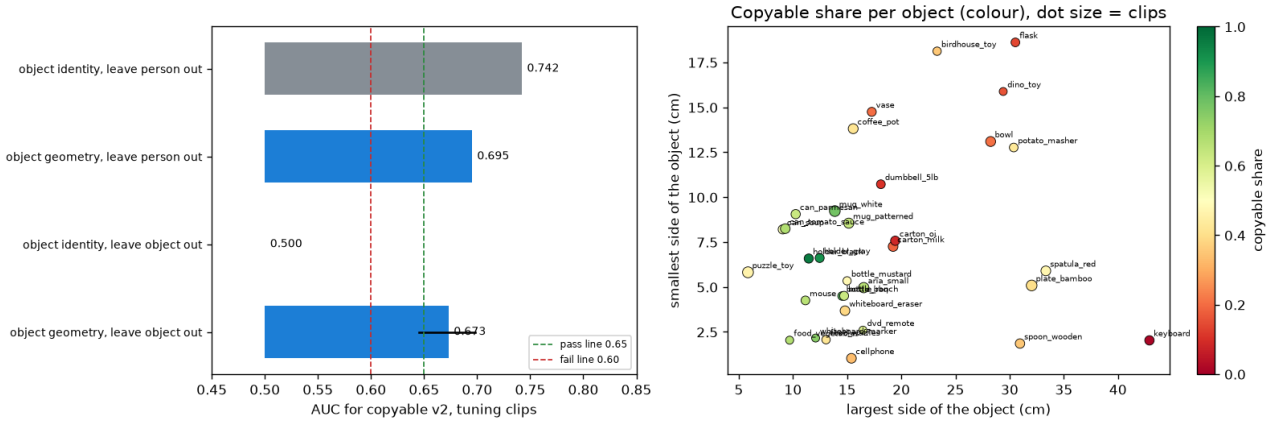


Figure 4: Object geometry from the 3D models against object identity, on tuning clips (EXP-18). Geometry still works when the object is left out of training, where identity cannot work at all.

Table 4: Comparison 5, what the checklist adds and what learning adds. Muse Spark grades only, paired over the 51 held-out sequences (EXP-40).

Row or difference	Value	95% interval	Gain
R1 Muse Spark alone	0.501	0.469 to 0.529	
R2 checklist, equal weights	0.617	0.579 to 0.659	
R3 checklist, learned	0.622	0.588 to 0.661	
R2 − R1 (the checklist)	+0.117	0.073 to 0.166	yes
R3 − R1	+0.122	0.081 to 0.166	yes
R3 − R2 (learning)	+0.005	−0.019 to 0.032	no
Ours − R3 (other checks)	+0.001	−0.011 to 0.013	no

Against collision failures, which are the most common, 0.611.

Learned weights. The largest weights are object shape (1.252), object width (0.922), finger occlusion (0.481), hand shape fit (0.434) and object size (0.391). Camera stability and visibility end up near zero, the first because almost no clip in HOT3D has an unstable camera, the second because visibility has only two effective grades.

7.4 How much the depth limit matters

The plan said to report the held-out result at 2 mm and 10 mm as well, and to learn nothing again. Table 5 and Figure 6 give it. At 10 mm the picture holds: our score 0.632, Muse Spark alone 0.541, the checklist adds, learning does not. At 2 mm, where only 20 percent of clips are copyable, our score falls to chance (0.521, the interval covers 0.5) and Muse Spark alone ranks clips the wrong way round (0.417). Our score still beats Muse Spark alone there, but only because Muse Spark alone is below chance. The object name reference stays at 0.76 to 0.77 at every depth.

Table 5: Task 2 held-out AUC at three collision depth limits. Nothing is learned again: all weights come from the 5 mm label on the 400 tuning clips, and only the held-out label being judged changes (EXP-41).

Row	2 mm	5 mm	10 mm
Held-out copyable share	19.9%	46.4%	64.7%
Our score	0.521	**0.624**	0.632
Muse Spark alone	0.417	0.501	0.541
Object name reference	0.761	0.771	0.774
Checklist, equal weights	0.488	0.617	0.624
Checklist, learned	0.506	0.622	0.629
Our score, unseen objects	0.504	0.571	0.585
Our score inside an object	0.505	0.506	0.504

So the task 2 result holds for the 5 mm limit that was fixed in advance and for a looser one, and it does not hold for a stricter one. We report this as a limit of the result, not as a choice between labels.

8 Soft objects: the check does not meet its rule

The score is geometric, and it is meant for rigid objects. A soft object changes shape in the hand, so its size and shape do not settle the question. The proposal therefore added one Muse Spark question per clip: is this object rigid, squeezable, soft or unsure? Soft or unsure means no score and a note that the clip needs a touch check.

Ground truth and rule, both fixed before any call. Two team members labeled all 104 object names rigid, squeezable or soft on their own, then agreed a single label. Agreement before settling was 81.7 percent with Cohen's kappa 0.613. The pass rule was at least 70 percent of soft segments held

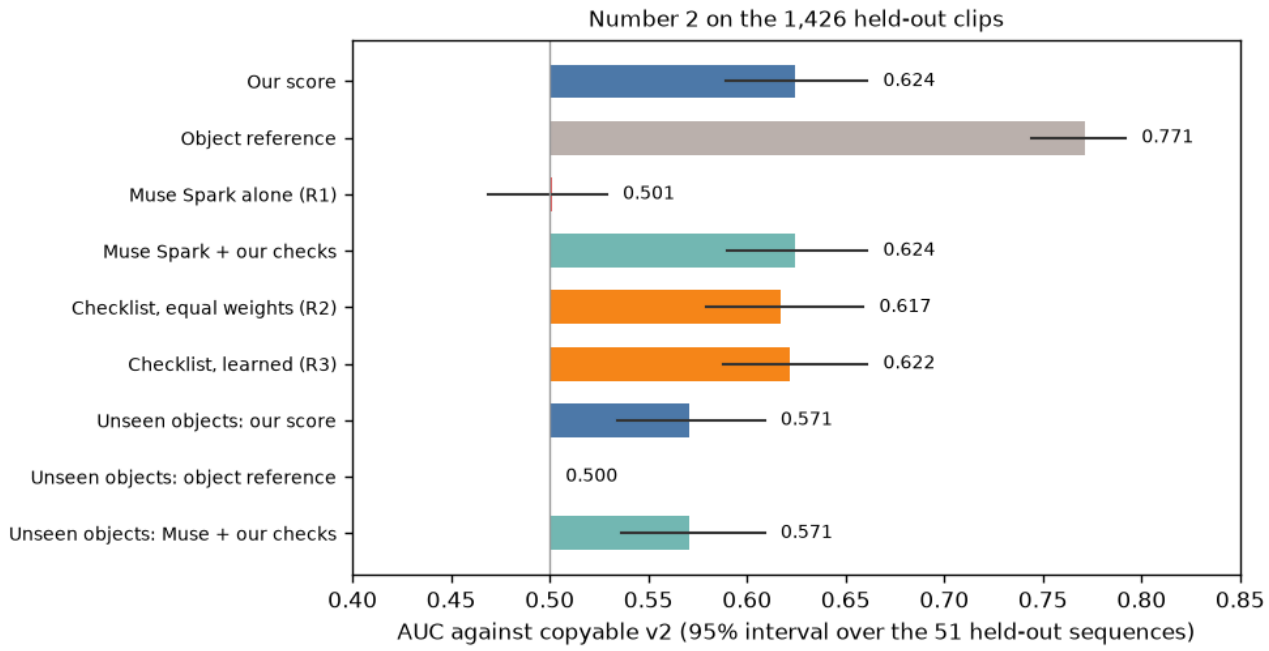


Figure 5: Task 2 on the 1,426 held-out clips (EXP-40).

Table 6: The soft object check against its pre-registered rule (EXP-42).

Metric	Value	Rule	Met
Soft segments held back	51.7%	$\geq 70\%$	no
Rigid HOT3D clips held back	2.1%	$\leq 10\%$	yes
Verdict: fail			

back and at most 10 percent of rigid HOT3D clips held back. HOT3D tracks every object as a rigid body, so any HOT3D clip held back counts as a mistake. There is no verdict with fewer than 20 soft segments or 4 soft objects.

The check fails (Table 6). All 1,782 items returned a parsed answer on the first pass. Muse Spark leaves rigid HOT3D clips alone, holding back only 2.1 percent of 1,426, which satisfies the second half of the rule. But it holds back only 31 of 60 segments labeled soft. The misses are almost all tape: the three tape names carry 32 of the 60 soft segments and only 28 percent of those are held back, while the other six soft objects are held back in 22 of 28 segments. Bubble wrap, gloves and mats are mostly caught. The check also holds back 18.1 percent of rigid OpenGraph segments, which the rule did not measure.

Two notes. Tape is a hard case for the labelers too: for two of the three tape names the two labelers disagreed before settling, one saying squeezable and the other soft. And dropping the held-back HOT3D clips leaves the task 2 AUC unchanged at 0.625, so the failed step does not damage the main result. It does not do the job it was added for. A "soft or unsure" answer from a vision language model is not enough to keep soft objects out of a geometric score.

9 The ground truth as a benchmark

Our heaviest and most checked work is the ground truth, not the checklist. We release both, labels only, with standalone scorers that need nothing from this project.

- `number1/ground_truth.json`: per episode and hand, contact segments from the cleaned glove and from the raw glove, as frame ranges at 30 fps. 50 tuning episodes with 955 contacts and 22 held-out episodes with 287.

- `number2/clips.csv`: per clip, where it is, the copyable label, the first condition that fails, and the label at the 2 mm and 10 mm depth limits. 2,385 tuning clips and 1,426 held-out clips over 33 objects.

The two scorers were written apart from the project code and need only numpy, pandas and scikit-learn. They reproduce the project numbers exactly on every case we checked, including boundary F1, contact IoU and the object share reference AUC with its bootstrap interval. No video, sensor value or motion capture pose is redistributed. Task 1 labels can be shared under CC BY 4.0. Task 2 labels derive from HOT3D hand data, so they go out under CC BY-NC-SA 4.0 and may not be sold. This is our reading of the license texts and not legal advice.

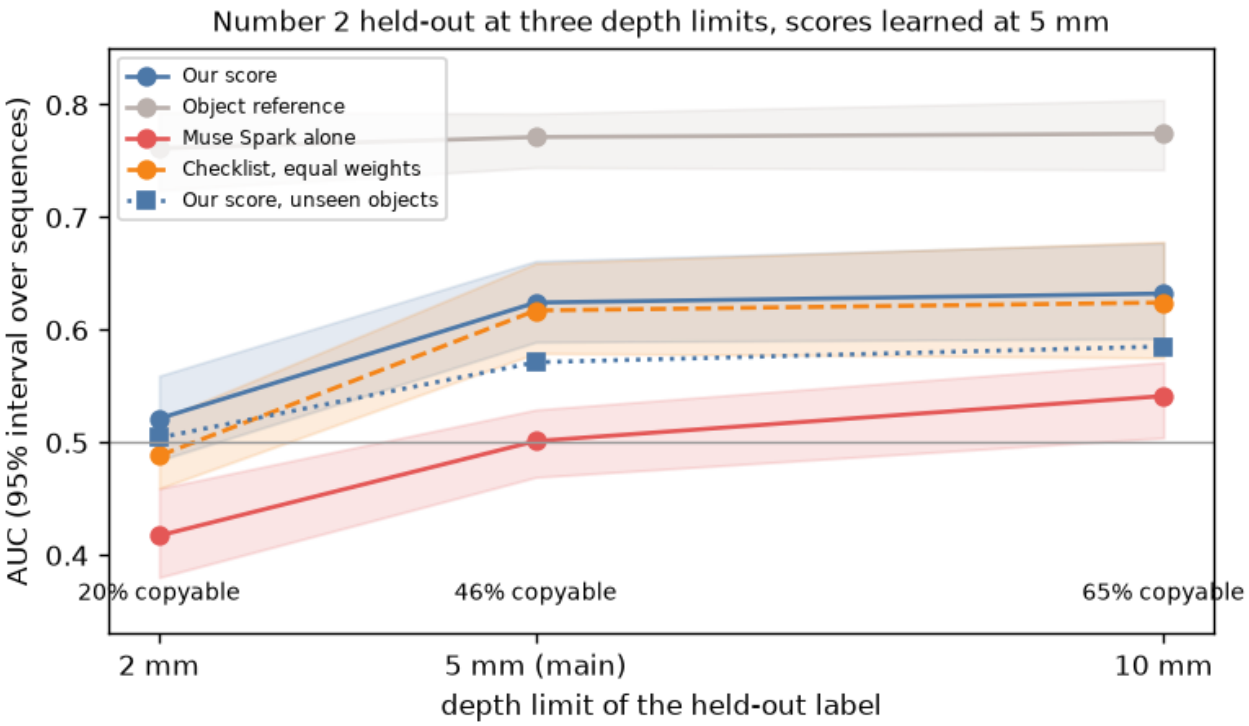


Figure 6: Task 2 held-out at three collision depth limits (EXP-41).

Table 7: Meta Model API spend, $143.27 of a $150 credit.

Step	Cost
Pilots, prompt checks and tuning checks	$19.06
Task 1: SAM on 50 tuning episodes	$11.89
Task 1: SAM on 22 held-out episodes	$5.45
Task 1: arm A on held-out	$12.44
Task 1: arm C, tuning sample and held-out	$30.28
Task 2: SAM on all clip windows	$26.48
Task 2: Muse Spark checklist and baseline	$28.56
Soft object check	$9.11
Total	**$143.27**

10 Cost

Every paid call is cached, so a rerun sends only what is missing, and all reported numbers can be recomputed with no API call.

Two choices did most of the saving. SAM at 10 fps gives the same ids and masks as at 30 fps, with a mask IoU of 0.97 to 0.99 on hands and the grasped object, which cuts SAM cost to a third (EXP-05). Overlapping clips of one sequence share one SAM window, so the task 2 SAM bill covered 1,826 clips with 513 windows for $26.48. SAM on all 2,152 usable grasp clips as single clips would have cost about $59 (EXP-07, EXP-11).

11 Limitations

- The robot check is geometric, not a physics test, and it uses one robot hand, the Allegro. It says that the hand can reach and hold the grasp without going into the object. It does not promise that the robot succeeds.
- The task 2 result depends on the collision depth limit. It holds at the 5 mm limit fixed in advance and at 10 mm, and not at 2 mm (Section 7.4).
- The object name alone predicts copyability better than our score when the object has labels, and our score does not rank two grasps of the same object (AUC 0.506 inside an object).
- The soft object step does not pass its rule, so soft objects are not reliably kept out of the score.
- Contact timing is low at ± 5 frames for every arm, because the glove measures pressure and video cannot see pressure. The strong rows are at ± 15 frames and in contact IoU.
- The HOT3D images are fisheye and stay raw. Undistorting them made MediaPipe worse, dropping hand detection from 70.6 to 60.3 percent, so a rule set in advance kept the raw frames.
- MediaPipe finds gloved hands poorly on the OpenGraph videos, from 14 percent to 100 percent of frames

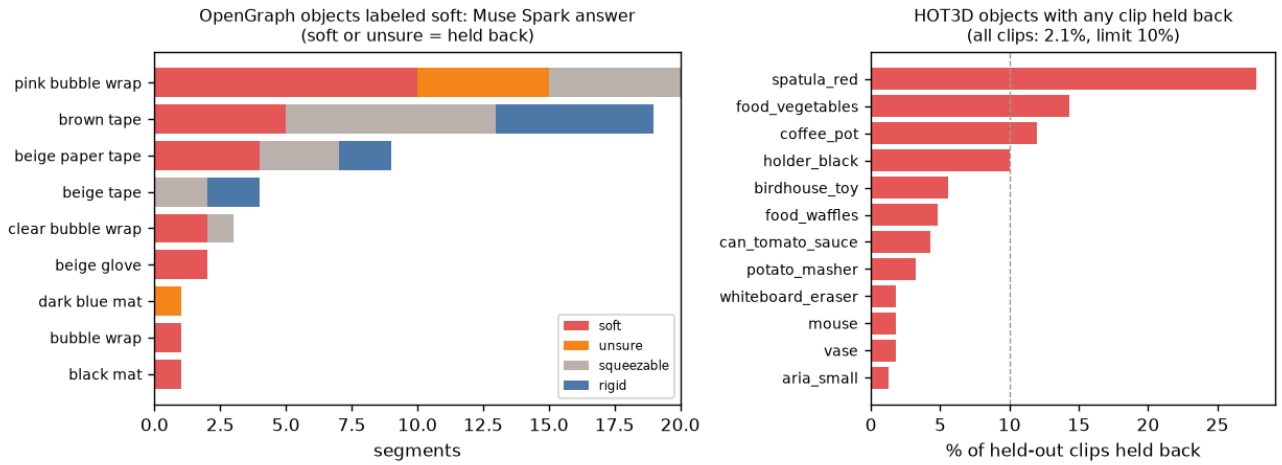


Figure 7: Muse Spark answers on the soft OpenGraph objects, and the HOT3D objects it holds back (EXP-42).

depending on the background. Task 1 therefore uses SAM and not MediaPipe, and task 2 runs on HOT3D, where hands are bare.

- The held-out set for task 2 has three persons, and the clips come from one glove setup and one motion capture setup. Everything here is lab data.

12 Conclusion

Two results hold on held-out data.

For contact timing, a vision language model asked directly about touch, plus one constant shift learned from a touch glove, beats a tracking rule built on segmentation masks. The reason is that the tracking rule cuts about three times too often. The shift is worth $+0.121$ F1 at ± 5 frames and it costs two numbers.

For robot copyability, asking fixed questions is worth more than training weights on labels. The plain question is at chance, 0.501. The same grades averaged with no learning reach 0.617, and learning the weights adds 0.005. The object name alone reaches 0.771 where the object has been measured before, so a filter can look strong by recognising the object and nothing else, and a single pooled AUC does not show this. On objects never seen in training, where the object name gives 0.5, the checklist keeps 0.571.

What did not work is reported as well. The soft object step misses its rule, the score cannot rank two grasps of the same object, and the main result does not hold at a stricter collision limit. The labels and the scorers are released so that anyone can check this, and score their own filter against the same sensors.

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